Machine Learning Case Study

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Problem Statement: Customer Purchase Predictions using Logistics Regression

Background:

In the business and retail sector, predicting customer purchase behaviour is essential for improving sales and marketing strategies. Businesses use customer data to identify buying patterns and predict purchase decisions. This project applies Logistic Regression to a Kaggle dataset containing 1,500 customer records to predict whether a customer will make a purchase.

Importance of the Problem:

Customer purchase prediction helps businesses understand their customers and improve marketing effectiveness.

- Identifies potential buyers.
- Supports targeted advertising.
- Reduces unnecessary marketing expenses.
- Improves customer engagement and sales planning.

Objective:

The main objective is to develop a Logistic Regression model that predicts whether a customer will purchase a product.

The project also aims to evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.

Challenges:

- Class Imbalance: Differences in purchase and non-purchase records may cause the model to favour the majority class.
- Complex Customer Behaviour: Purchase decisions depend on multiple factors, such as income, age, discounts, and shopping habits.

- Prediction Errors: False positives and false negatives can lead to ineffective marketing strategies and missed potential customers.
- Feature Dependence: The model's performance depends on the quality and relevance of the available customer data.

Applications:

Customer purchase prediction can be applied in:

- E-commerce: Identifying potential buyers.
- Digital marketing: Creating targeted advertising campaigns.
- Retail: Understanding customer buying patterns.
- Customer analytics: Supporting customer segmentation and sales planning.

Methodology:

Workflow and Strategy:

The project follows a systematic workflow:

1. Upload the Kaggle dataset into Google Colab.
2. Explore the dataset and check for missing values.
3. Separate input features from the target variable.
4. Split the data into 80% training and 20% testing sets.
5. Standardize the features using `StandardScaler`.
6. Train the Logistic Regression model.
7. Predict customer purchase status.
8. Evaluate and visualize the results.

Implementation:

- **Dataset Name:** Customer Purchase Dataset (https://www.kaggle.com/datasets/rabieelkharoua/predict-customer-purchase-behavior-dataset/data)
- **Target Variable:** `PurchaseStatus` (0 = No Purchase, 1 = Purchase)
- **Key Features:** Customer demographic details, income, shopping behaviour, and other customer-related attributes used to predict purchase decisions.

Python link:

https://colab.research.google.com/drive/1YU1MFOmhG0mLxwpkaZKxA0h5wOB2m0F9?usp=sharing

Github Link:

Results Analysis and Display

The Logistic Regression model achieved 82.67% accuracy on the test dataset of 300 records.

Evaluation metric	No Purchase	Purchase
Precision	0.83	0.82
Recall	0.88	0.76
F1-score	0.85	0.79

The model correctly classified 248 records and incorrectly classified 52 records.

The results indicate that the model identifies customers who do not purchase more effectively than customers who do purchase. The confusion matrix and classification report provide further insight into its performance.

Strengths and Weaknesses Analysis:

- Logistic Regression:

- Strengths: Simple to implement, computationally efficient, and provides purchase probabilities for clear decision-making.

- Weaknesses: May struggle to capture complex, non-linear relationships in customer behaviour and can make incorrect predictions.

- Decision Tree:

- Strengths: Easy to understand, handles non-linear relationships, and identifies important factors influencing purchase decisions.
- Weaknesses: Can overfit the training data and may perform poorly on unseen customer records.

- Random Forest:

- Strengths: Combines multiple decision trees to improve prediction accuracy, handles complex customer patterns, and provides feature importance scores.
- Weaknesses: Requires more computational resources and is less interpretable than Logistic Regression.

Key Insights and Conclusions:

The project demonstrates that Logistic Regression can be used to predict customer purchase behaviour using machine learning. The model achieved 82.67% accuracy, showing that it correctly classified a substantial proportion of the test records.

The results highlight the importance of data preprocessing, model evaluation, and visualization. Although the model performed reasonably well, further improvements could be explored by testing additional features, tuning model parameters, and comparing it with algorithms such as Decision Trees and Random Forest.

Overall, the project provides a practical introduction to machine learning and its applications in customer analytics and targeted marketing.